Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

MT306S – VECTOR AND FOURIER ANALYSIS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. If $\vec{r} = 3t^2\hat{\imath} + (t^2 - 2t)\hat{\jmath} + t^3\hat{k}$, then find its velocity at time $t = 1$.
2. Find $\nabla.\vec{r}$.
3. State Green's theorem.
4. Find $f(x) = \frac{1}{2}(\pi - x), 0 < x < 2\pi$, find a_0 .
5. Write Parseval's identity for Fourier transform.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. If $\vec{r} = e^{-t}\hat{\imath} + 2cos3t\hat{\jmath} + 2sin3t\hat{k}$, then find its acceleration at $t = 0$.
7. Show that $\overrightarrow{F} = 3x^2y\hat{\imath} - 4xy^2\hat{\jmath} + 2xyz\hat{k}$ is solenoidal.
8. If $\vec{F} = 3xy\hat{\imath} - y^2\hat{\jmath}$, evaluate $\int_C \vec{F}.d\vec{r}$ where C is the curve on the xy plane $y = 2x^2$ from $(0,0)$ to $(1,2)$.
9. Show that for all values of x on $(-\pi, \pi)$, $\frac{x}{2} = sinx - \frac{sin2x}{2} + \frac{sin3x}{3} - \frac{sin4x}{4} + \cdots\infty$.
10. Obtain the Fourier cosine transform for e^{-ax}, $a > 0$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) For the curve $\vec{r} = e^{-t}\vec{\imath} + \log(t^2 + 1)\vec{\jmath} - tan t\vec{k}$, find the velocity and acceleration at $t = 0$.

(or)

b) If $\vec{a}, \vec{b}, \vec{\omega}$ are vector function of a scalar variable u and if

$\frac{d\vec{a}}{du} = (\vec{\omega} \times \vec{a})$, $\frac{d\vec{b}}{du} = (\vec{\omega} \times \vec{b})$, $\frac{d}{du}(\vec{a} \times \vec{b}) = \vec{\omega} \times (\vec{a} \times \vec{b})$

12. a) i) If $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, Show that $\nabla(r^n) = nr^{n-2}\,\vec{r}$.
ii) If $\vec{F} = xyz^2\hat{\jmath} + x^2yz\hat{k}$, find div curl $\vec{F}$.

(or)

b) i) If $\vec{A} = (axy - z^3)\hat{\imath} + (a-2)x^2\hat{\jmath} + (1-a)xz^2\hat{k}$ is irrotational, find a.
ii) Show that $\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla.\vec{F}) - \nabla^2\vec{F}$ where $\vec{F}$ is a vector point function.

13. a) If $\vec{F} = 4xz\hat{\imath} - y^2\hat{\jmath} + yz\hat{k}$, evaluate $\iint_s \vec{F}.\hat{n}\,ds$ where s is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

(or)

b) Verify Green's theorem in the plane for $\oint_c (xy + y^2)\,dx + x^2dy$, where C is the chord of the region bounded by $y = x \; and \; y = x^2$.

14. a) Obtain the Fourier series for the function $f(x) = x^2$ in the interval $-\pi < x < \pi$ and hence Show that $\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \infty = \frac{\pi^2}{6}$.

(or)

b) Obtain the cosine series for $f(x) = \begin{cases} cosx, & 0 < x < \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < x < \pi \end{cases}$

15. a) Find the Fourier Sine transform of $f(x) = e^{-x}cosx$.

(or)

b) Use Parseval's identity to Show that $\int_0^\infty \frac{1}{(x^2+a^2)(x^2+b^2)}\,dx = \frac{\pi}{2ab(a+b)}$.

****